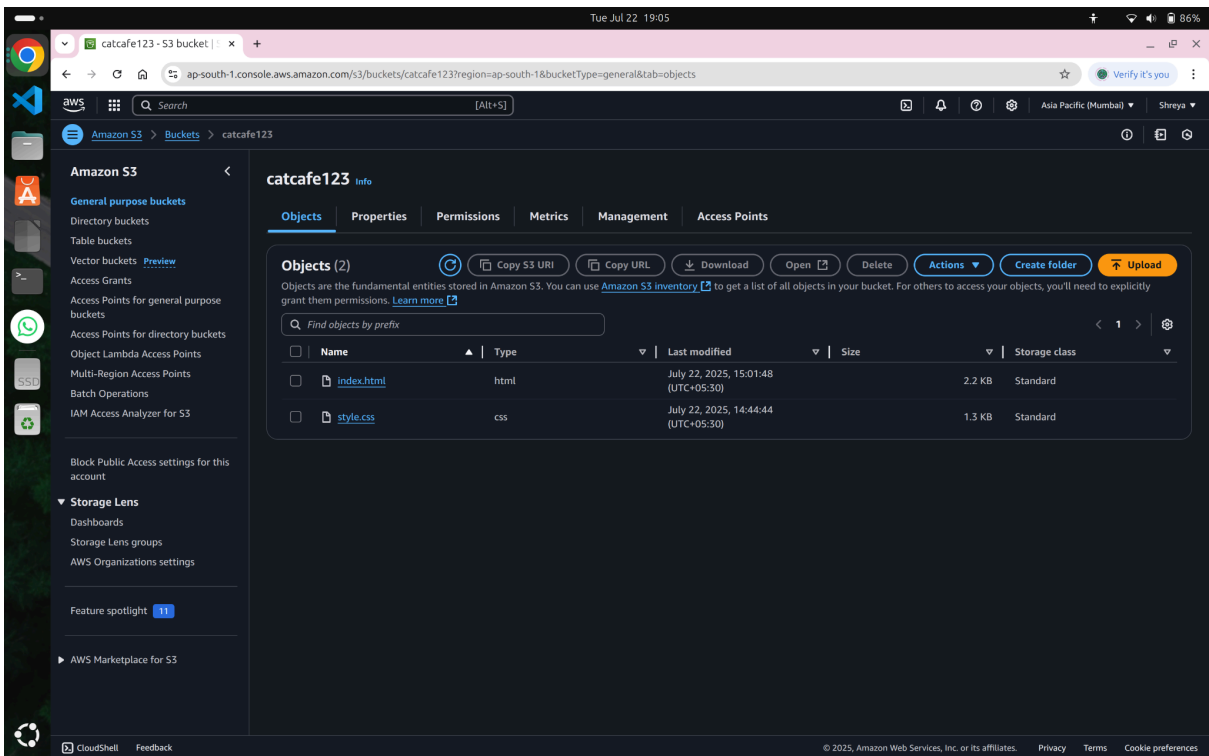
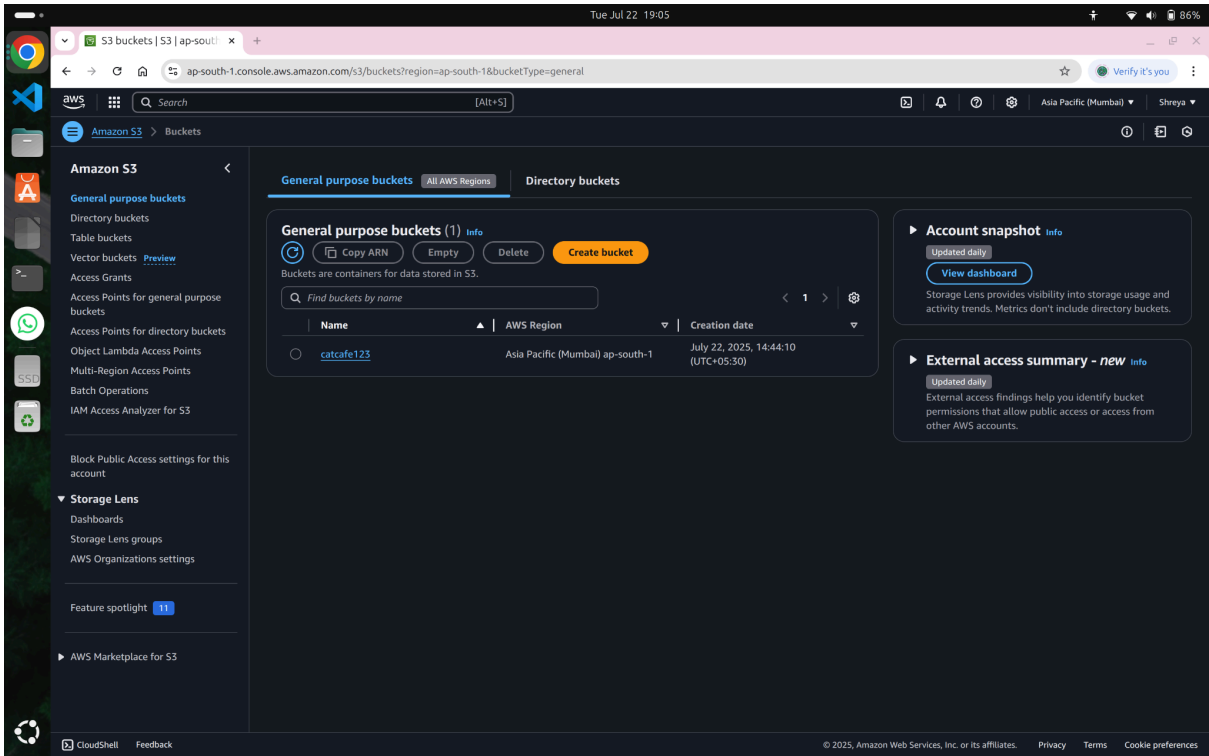




catcafe123 - S3 bucket

ap-south-1.console.aws.amazon.com/s3/buckets/catcafe123?region=ap-south-1&bucketType=general&tab=properties

Verify it's you

Search

[Alt+S]

Asia Pacific (Mumbai) Shreya

Amazon S3

Buckets

catcafe123

General purpose buckets

Directory buckets

Table buckets

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catcafe123

Info

Objects

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Permissions

Metrics

Management

Access Points

Bucket overview

AWS Region

Amazon Resource Name (ARN)

Creation date

Bucket Versioning

Multi-factor authentication (MFA) delete

Tags (0)

Default encryption

Asia Pacific (Mumbai) ap-south-1

arn:aws:s3:::catcafe123

July 22, 2025, 14:44:10 (UTC+05:30)

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures.

Disabled

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API.

Disabled

You can use bucket tags to track storage costs and organize buckets.

No tags associated with this resource.

Server-side encryption is automatically applied to new objects stored in this bucket.

CloudShell

Feedback

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catcafe123 - S3 bucket

ap-south-1.console.aws.amazon.com/s3/buckets/catcafe123?region=ap-south-1&bucketType=general&tab=properties

Verify it's you

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Asia Pacific (Mumbai) Shreya

Amazon S3

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Object Lock

Requester pays

Static website hosting

Disabled

Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets.

Disabled

When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled.

Disabled

Use this bucket to host a website or redirect requests.

We recommend using AWS Amplify Hosting for static website hosting

S3 static website hosting

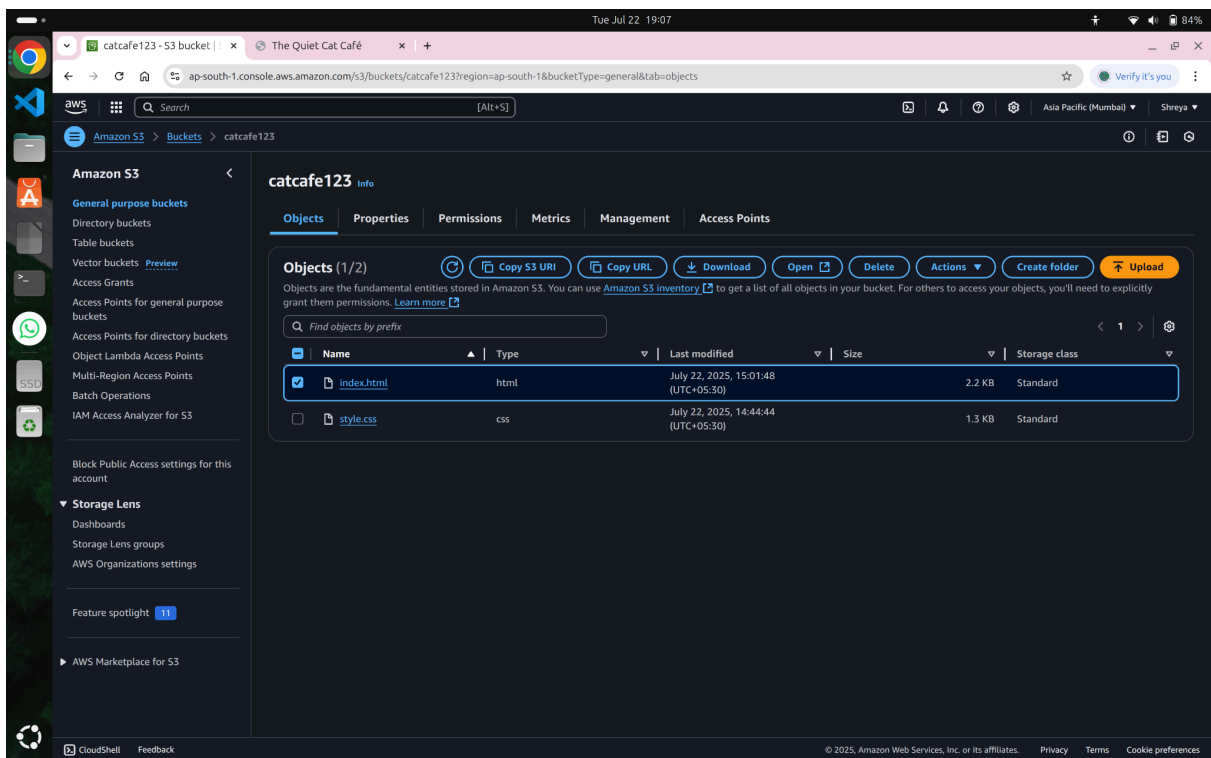
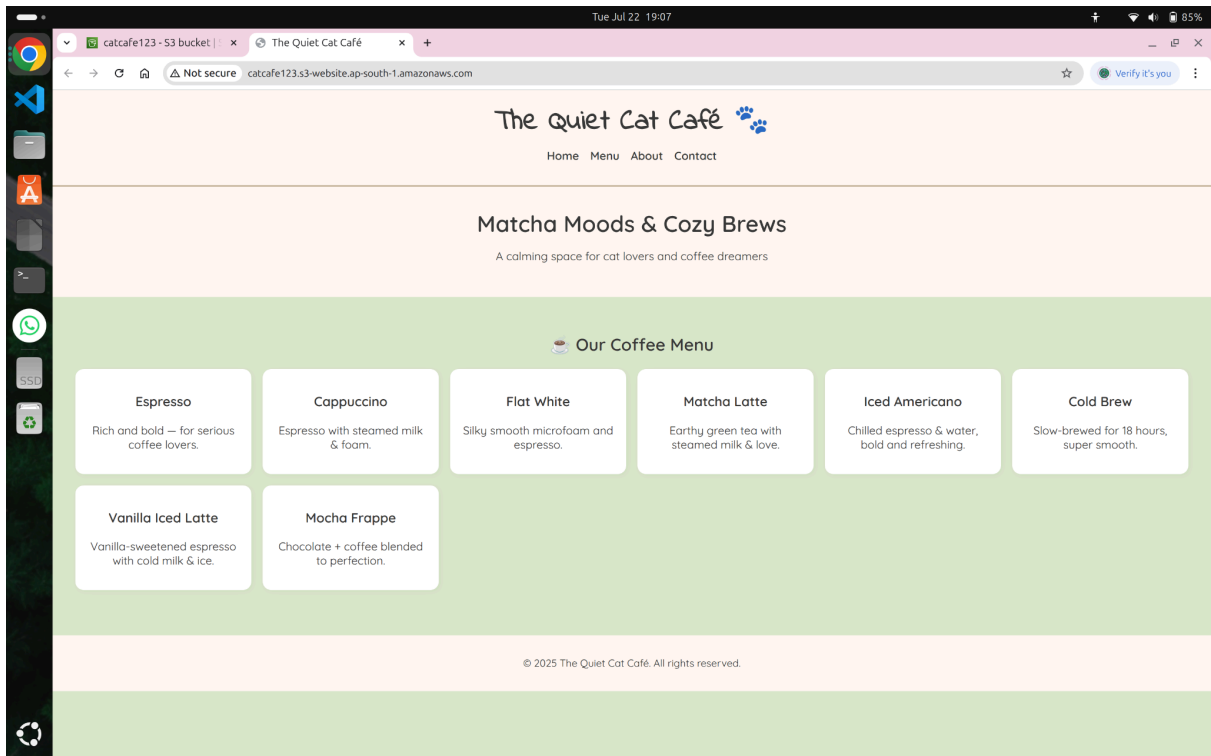
Hosting type

Bucket website endpoint

CloudShell

Feedback

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The commands to use the AWS CLI (and my entire progression): Microsoft Windows
 [Version 10.0.26100.4770]
 (c) Microsoft Corporation. All rights reserved.

```
C:\Users\shrey>aws --version
aws-cli/2.27.60 Python/3.13.4 Windows/11 exe/AMD64
```

```
C:\Users\shrey>aws configure
AWS Access Key ID [None]: [REDACTED]
AWS Secret Access Key [None]: [REDACTED]
Default region name [None]: ap-south-1
Default output format [None]: json
```

```
C:\Users\shrey>aws s3 mb s3://shreyas-matcha-site
make_bucket: shreyas-matcha-site
```

```
C:\Users\shrey>aws s3 cp
"C:\Users\shrey\OneDrive\Desktop\Programming\HTML\index.html"
s3://shreyas-matcha-site
upload: OneDrive\Desktop\Programming\HTML\index.html to
s3://shreyas-matcha-site/index.html
```

```
C:\Users\shrey>aws s3api put-object-acl --bucket shreyas-matcha-site --key index.html --acl
public-read
```

An error occurred (AccessDenied) when calling the PutObjectAcl operation: User: arn:aws:iam::536784880307:root is not authorized to perform: s3:PutObjectAcl on resource: "arn:aws:s3:::shreyas-matcha-site/index.html" because public access control lists (ACLs) are blocked by the BlockPublicAcls block public access setting.

```
C:\Users\shrey>aws s3api put-public-access-block \
```

usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]

To see help text, you can run:

```
aws help
aws <command> help
aws <command> <subcommand> help
```

aws: error: the following arguments are required: --bucket,
--public-access-block-configuration

```
C:\Users\shrey> --bucket shreyas-matcha-site \
'--bucket' is not recognized as an internal or external command,
operable program or batch file.
```

```
C:\Users\shrey> --public-access-block-configuration
"BlockPublicAcls=false,IgnorePublicAcls=false,BlockPublicPolicy=false,RestrictPublicBucket
s=false"
'--public-access-block-configuration' is not recognized as an internal or external command,
operable program or batch file.
```

```
C:\Users\shrey>aws s3api put-public-access-block --bucket shreyas-matcha-site
--public-access-block-configuration
BlockPublicAcls=false,IgnorePublicAcls=false,BlockPublicPolicy=false,RestrictPublicBuckets
=false
```

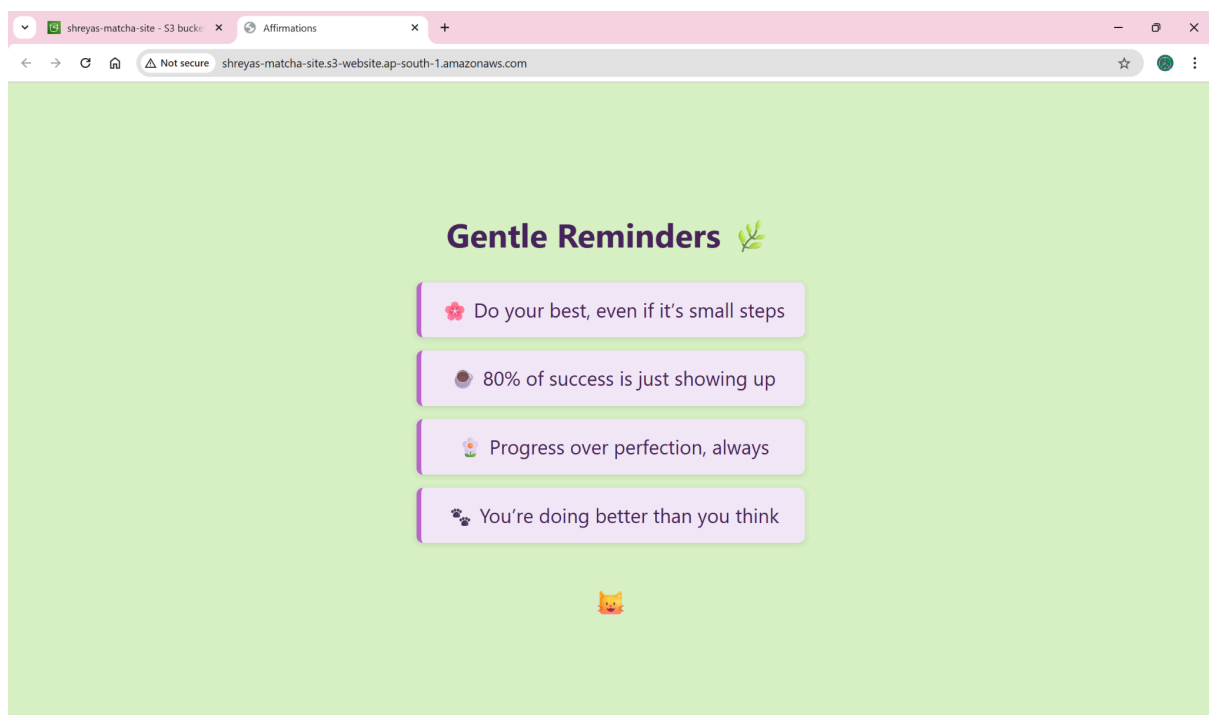
```
C:\Users\shrey>aws s3api put-object-acl --bucket shreyas-matcha-site --key index.html --acl
public-read
```

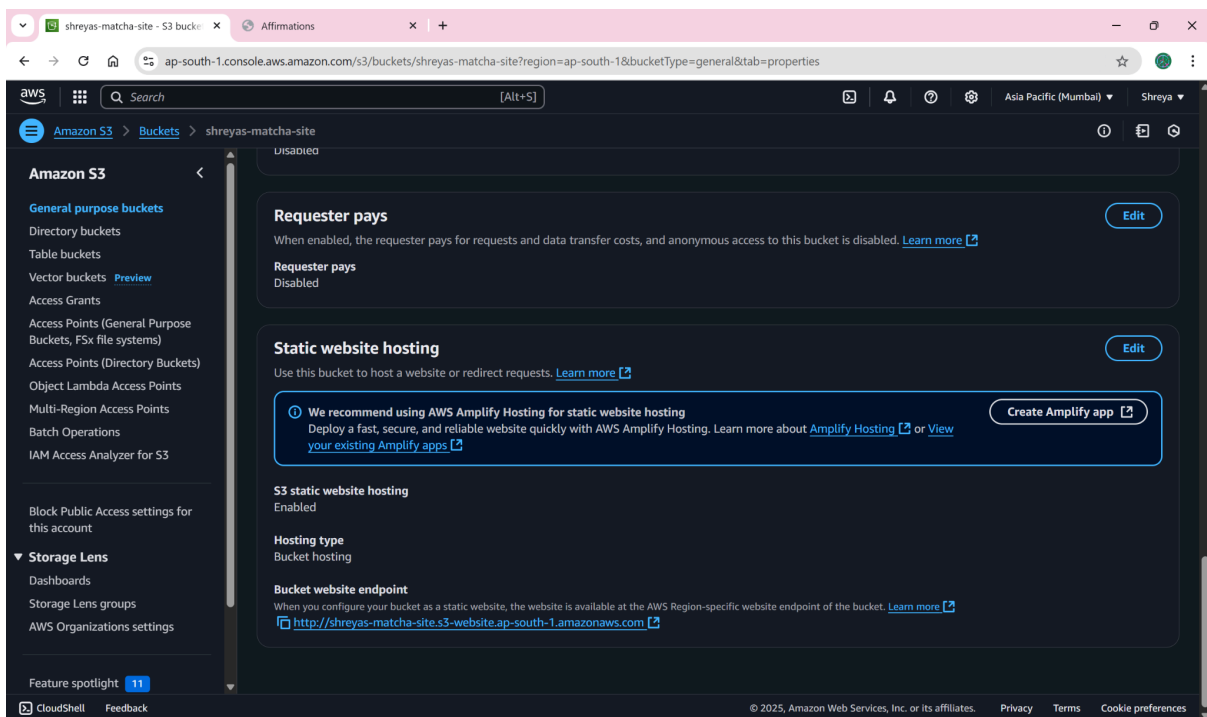
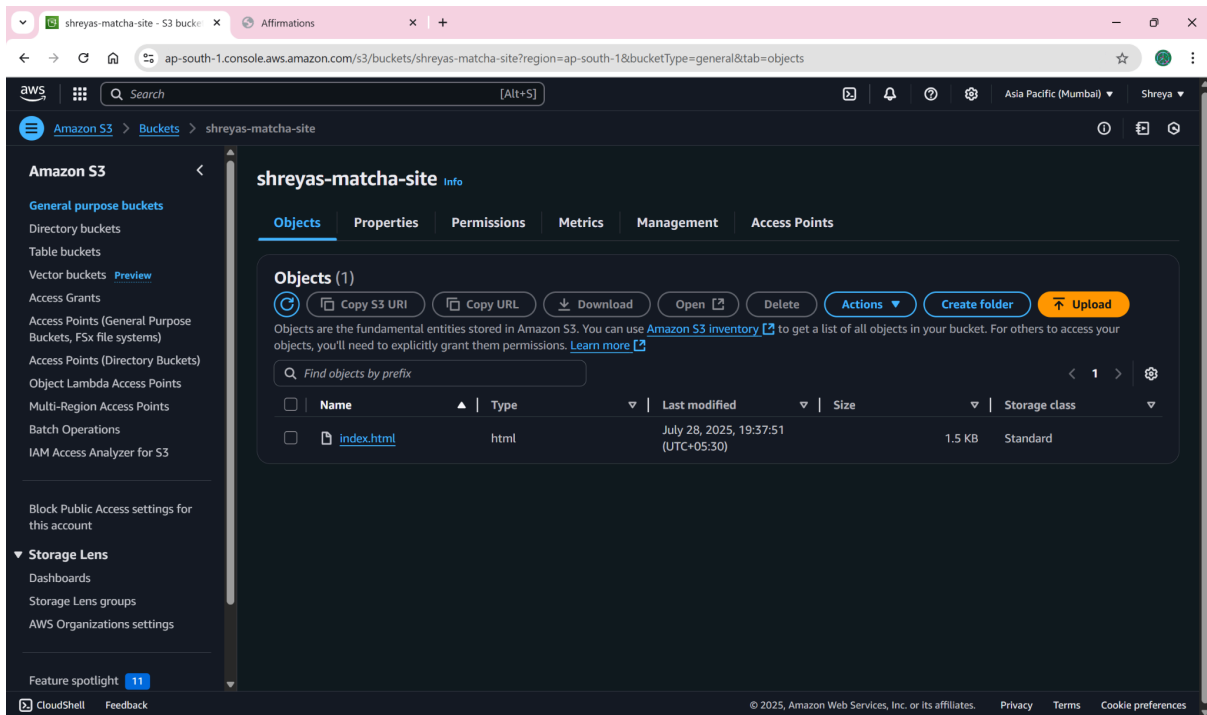
An error occurred (AccessControlListNotSupported) when calling the PutObjectAcl operation: The bucket does not allow ACLs

```
C:\Users\shrey>aws s3api put-bucket-policy --bucket shreyas-matcha-site --policy
file://policy.json
```

```
C:\Users\shrey>aws s3 cp
"C:\Users\shrey\OneDrive\Desktop\Programming\HTML\index.html"
s3://shreyas-matcha-site
upload: OneDrive\Desktop\Programming\HTML\index.html to
s3://shreyas-matcha-site/index.html
```

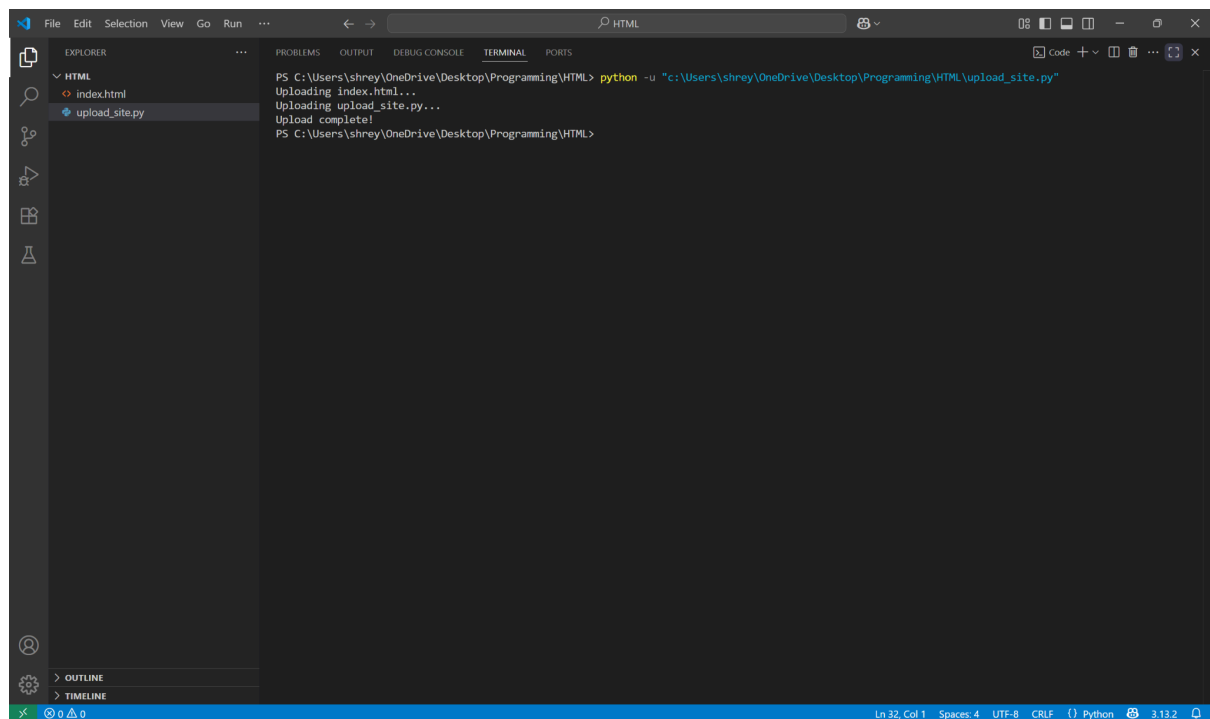
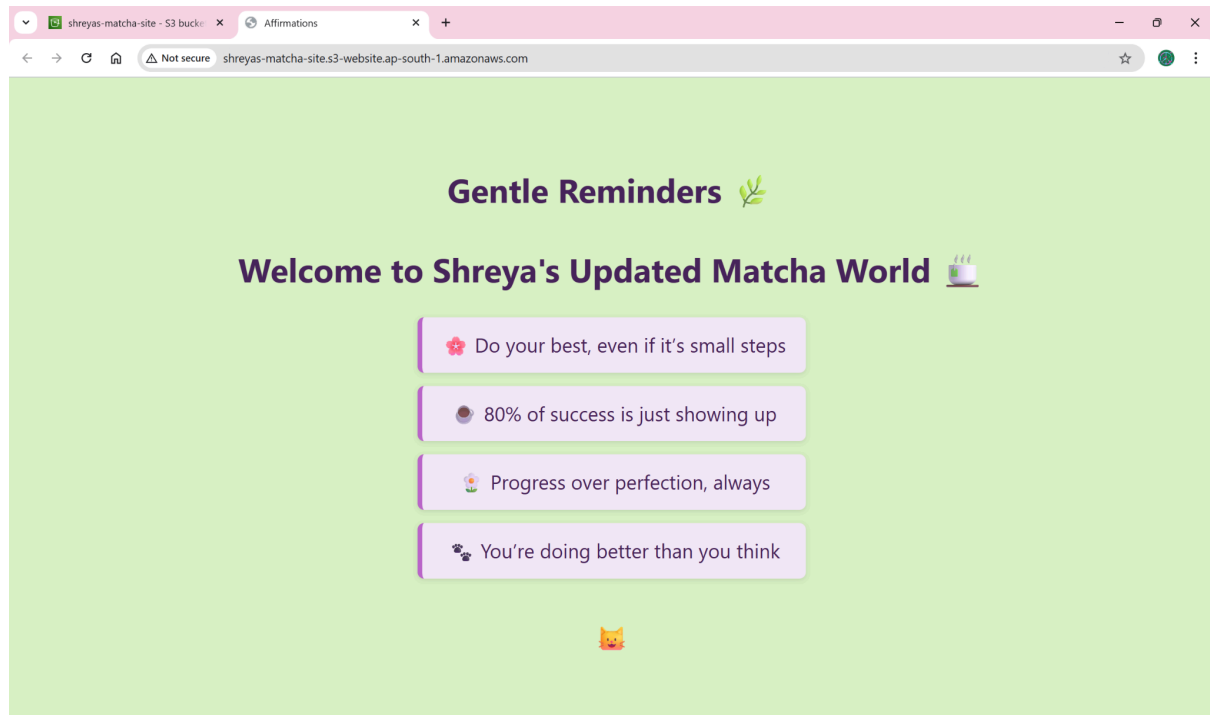
```
C:\Users\shrey>
```





Now using the Python code to automate it:

Output :



The code:

```
import subprocess
import os

# Configuration
region = "ap-south-1"
```

```

bucket_name = "shrey-static-site-bucket-001"
local_directory = r"C:/Users/shrey/OneDrive/Desktop/Programming/HTML"
# local path to site files
download_path = r"C:/Users/shrey/Downloads/downloaded-index.html"
backup_folder = r"C:/Users/shrey/Downloads/s3-backup"

# --- Helper to run CLI commands ---
def run_command(description, command):
    print(f"\n✅ {description}")
    print(f"▶ Running: {command}")
    try:
        result = subprocess.run(command, shell=True, check=True,
text=True, capture_output=True)
        print(result.stdout)
        return True
    except subprocess.CalledProcessError as e:
        print(f"❌ Error:", e.stderr)
        return False

# --- 1. Create Bucket ---
create_bucket_cmd = f'aws s3api create-bucket --bucket {bucket_name}
--region {region} --create-bucket-configuration
LocationConstraint={region}'
run_command("Create Bucket", create_bucket_cmd)

# --- 2. List Buckets ---
list_buckets_cmd = 'aws s3api list-buckets'
run_command("List Buckets", list_buckets_cmd)

# --- 3. Upload Directory ---
print(f"\n✅ Uploading all files from {local_directory} to S3
bucket...")
upload_directory_cmd = f'aws s3 cp "{local_directory}"/"
s3://{bucket_name}/ --recursive'
if run_command("Upload Directory", upload_directory_cmd):
    print("Upload complete!")

# --- 4. List Objects ---
list_objects_cmd = f'aws s3 ls s3://{bucket_name}/ --recursive'
run_command("List All Objects in Bucket", list_objects_cmd)

# --- 5. Download Single File (example with index.html) ---
download_cmd = f'aws s3 cp s3://{bucket_name}/index.html
"{download_path}"'
run_command("Download Single File", download_cmd)

# --- 6. Backup Bucket (Download All Files) ---
try:
    os.makedirs(backup_folder, exist_ok=True)
    backup_cmd = f'aws s3 cp s3://{bucket_name}/ "{backup_folder}"/"
--recursive'

```



```

    run_command("Backup All Files from Bucket", backup_cmd)
except Exception as e:
    print(f"❌ Error creating backup folder: {e}")

# --- 7. Sync Directory (more efficient than cp for updates) ---
print(f"\n✅ Alternative: Sync local directory with S3 bucket (only uploads changed files)")
sync_cmd = f'aws s3 sync "{local_directory}" s3://{bucket_name}/'
run_command("Sync Directory to S3", sync_cmd)

# --- 8. Restore from Backup ---
restore_cmd = f'aws s3 cp "{backup_folder}" s3://{bucket_name}/ --recursive'
run_command("Restore Backup to Bucket", restore_cmd)

# --- 9. Delete Single File ---
delete_file_cmd = f'aws s3 rm s3://{bucket_name}/index.html'
run_command("Delete Single File from Bucket", delete_file_cmd)

# --- 10. Empty and Delete Bucket ---
empty_bucket_cmd = f'aws s3 rm s3://{bucket_name} --recursive'
run_command("Empty Bucket", empty_bucket_cmd)

delete_bucket_cmd = f'aws s3 rb s3://{bucket_name}'
run_command("Delete Bucket", delete_bucket_cmd)

# --- Additional Useful Commands ---
print(f"\n📋 Additional useful AWS CLI commands:")
print(f"• Check bucket size: aws s3 ls s3://{bucket_name}/ --recursive --human-readable --summarize")
print(f"• Sync with delete: aws s3 sync \"{local_directory}\" s3://{bucket_name}/ --delete")
print(f"• Set public read access: aws s3api put-bucket-acl --bucket {bucket_name} --acl public-read")
print(f"• Enable website hosting: aws s3 website s3://{bucket_name}/ --index-document index.html --error-document error.html")
print(f"• Check bucket policy: aws s3api get-bucket-policy --bucket {bucket_name}")
print(f"• Get bucket location: aws s3api get-bucket-location --bucket {bucket_name}")

# --- Optional: Interactive Menu ---
def interactive_menu():
    """Optional interactive menu for selective operations"""
    operations = {
        '1': ('Create Bucket', create_bucket_cmd),
        '2': ('Upload Directory', upload_directory_cmd),
        '3': ('List Objects', list_objects_cmd),
        '4': ('Download File', download_cmd),
        '5': ('Backup Bucket', backup_cmd),
        '6': ('Sync Directory', sync_cmd),
    }

```

```

        '7': ('Delete File', delete_file_cmd),
        '8': ('Empty Bucket', empty_bucket_cmd),
        '9': ('Delete Bucket', delete_bucket_cmd)
    }

    print("\n🔧 Interactive Mode - Choose an operation:")
    for key, (desc, _) in operations.items():
        print(f"{key}. {desc}")

    choice = input("\nEnter your choice (1-9) or 'q' to quit: ")

    if choice == 'q':
        return False
    elif choice in operations:
        desc, cmd = operations[choice]
        run_command(desc, cmd)
    else:
        print("❌ Invalid choice")

    return True

# Uncomment the following lines to enable interactive mode
print("\n" + "="*50)
print("🚀 AWS CLI S3 Automation Script Complete!")
print("="*50)
while interactive_menu():
    continue

```